

Title: Embedded IoT and Real-Time Systems

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- Year / Semester: 2nd year / 3rd semester

Abstract Embedded Internet of Things (IoT) and real-time systems are essential technologies that support intelligent and time-critical applications in modern engineering fields. Embedded IoT systems integrate microcontrollers, sensors, actuators, and communication modules to collect, process, and transmit data through the internet. Real-time systems ensure that tasks are executed within strict timing constraints, where system correctness depends on both logical accuracy and timely response. These systems are widely applied in areas such as industrial automation, healthcare monitoring, smart homes, automotive systems, and smart city infrastructure. The architecture of embedded IoT and real-time systems typically includes real-time operating systems (RTOS), scheduling mechanisms, and reliable communication protocols to achieve low latency, high reliability, and efficient resource utilization. **This abstract highlights** the core concepts, components, and challenges of embedded IoT and real-time systems, including power management, scalability, security, and real-time performance, emphasizing their importance in developing responsive and reliable smart applications.

Keywords: Embedded Systems, Internet of Things (IoT), Real-Time Systems, RTOS, Automation